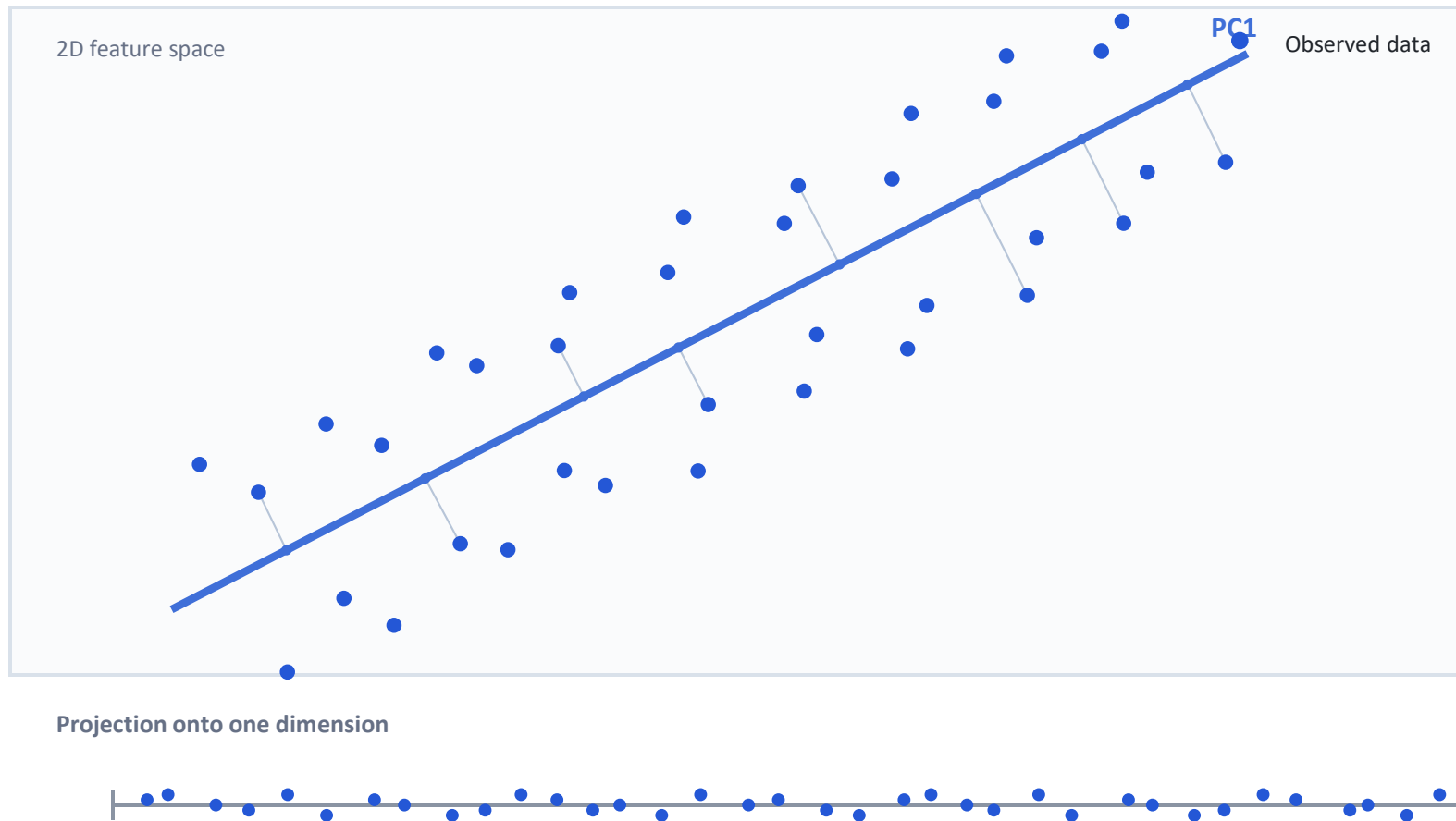


Lecture 4

Fisher's Linear Discriminant Analysis (Two class problem)

PCA follows the direction of maximum variance

1 UNLABELED DATA



PRINCIPAL COMPONENT

Objective

Maximize total variance

Labels

Not used

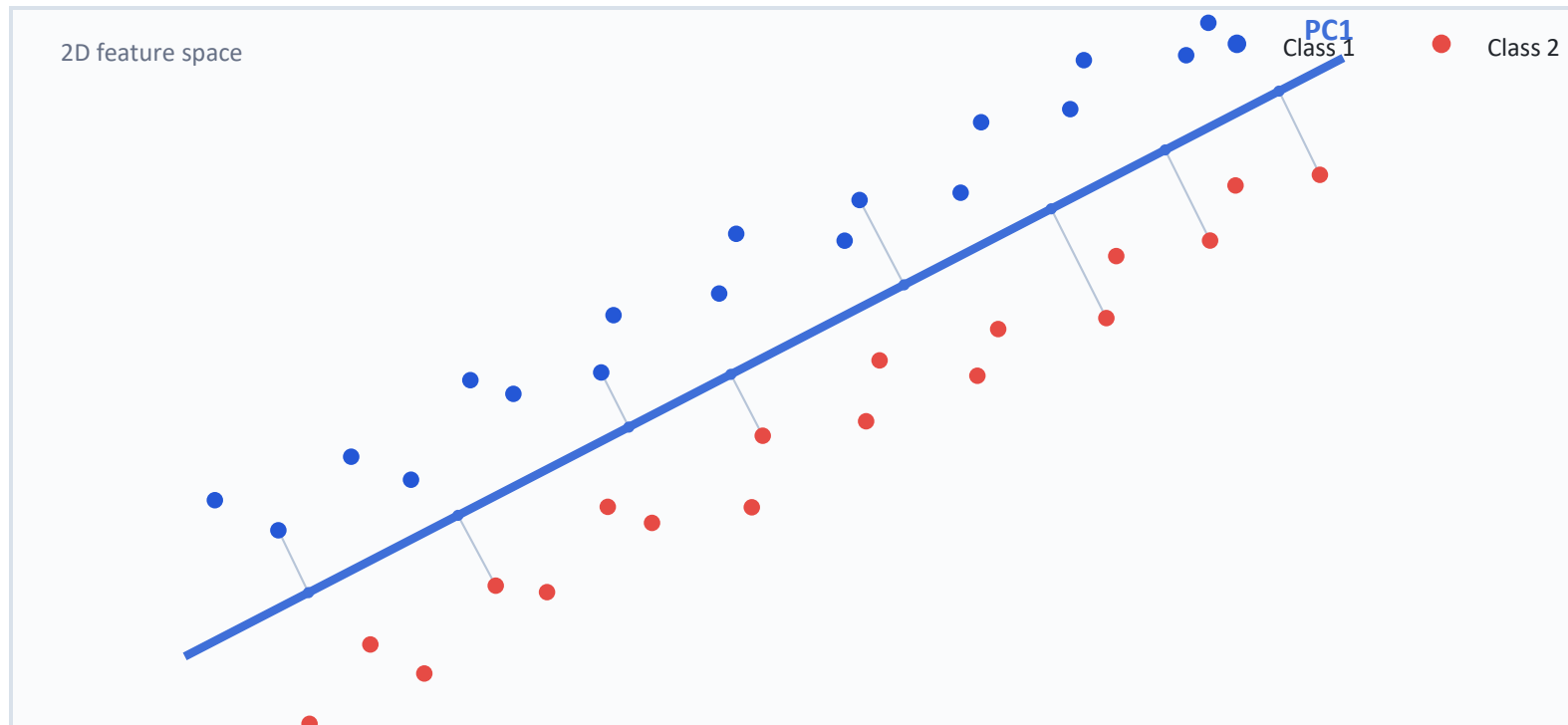
Result

PC1 follows the long direction of the cloud

PCA summarizes the overall geometry before any classes are known.

PCA projection mixes the red and blue classes

2 REVEAL THE CLASSES



Projection onto one dimension



PCA ignores labels, so the two classes overlap after projection.

PRINCIPAL COMPONENT

Same direction

Coloring the points does not change PC1

1D projection

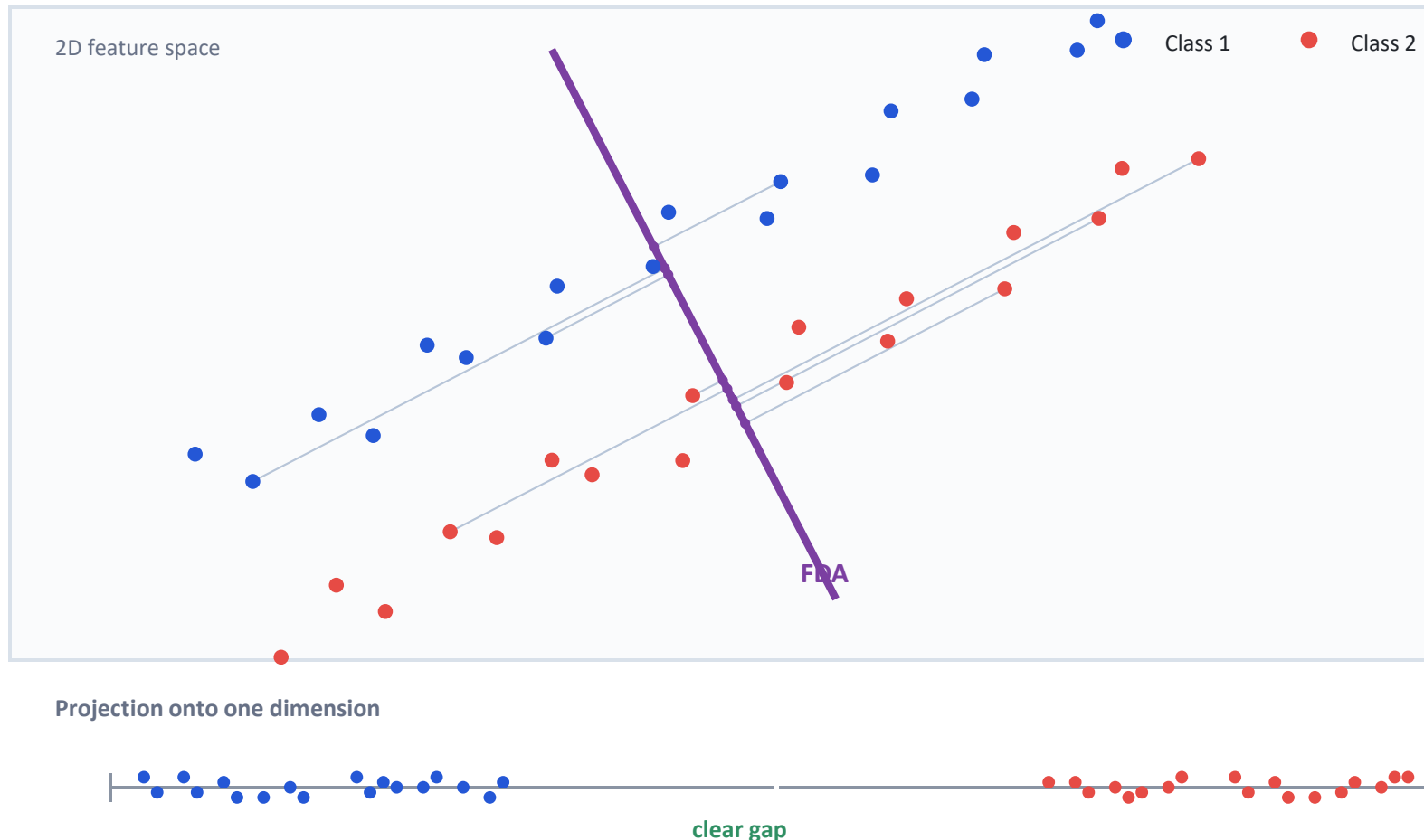
Red and blue occupy the same range

Why

Maximum variance is not the same as class separation

FDA projection cleanly separates the two classes

3 USE THE CLASS LABELS



FDA uses class labels to choose the direction that best separates them.

FISHER DISCRIMINANT

Objective

Maximize distance between class means

Control

Minimize spread within each class

1D projection

Red and blue form distinct groups

Fisher's Linear Discriminant Analysis (Two class problem)

- ▶ Assume we have only two classes.
- ▶ The idea behind Fisher's Linear Discriminant Analysis is to reduce the dimensionality of the data to one dimension. That is, to take d -dimensional $\mathbf{x} \in \mathbb{R}^d$ and map it to one dimension by finding $\mathbf{w}^T \mathbf{x}$:

$$z = \mathbf{w}^T \mathbf{x} = \begin{bmatrix} w_1 & \cdots & w_d \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_d \end{bmatrix} = \sum_{i=1}^d w_i x_i$$

- ▶ The one-dimensional z is then used for classification.

Fisher's Linear Discriminant Analysis

Goal: To find a direction such that projected data $\mathbf{w}^T \mathbf{x}$ are well separated.

Consider the two-class problem:

$$\mu_0 = \frac{1}{n_0} \sum_{i:y_i=0} x_i \quad \mu_1 = \frac{1}{n_1} \sum_{i:y_i=1} x_i$$

We want to:

1. Maximize the distance between projected class means.
2. Minimize the within class variance.

Fisher's Linear Discriminant Analysis

The distance between projected class means is:

$$\begin{aligned}(\mathbf{w}^T \mu_0 - \mathbf{w}^T \mu_1)^2 &= (\mathbf{w}^T \mu_0 - \mathbf{w}^T \mu_1)^T (\mathbf{w}^T \mu_0 - \mathbf{w}^T \mu_1) \\&= (\mu_0 - \mu_1)^T \mathbf{w} \mathbf{w}^T (\mu_0 - \mu_1) \\&= \mathbf{w}^T (\mu_0 - \mu_1) (\mu_0 - \mu_1)^T \mathbf{w} \\&= \mathbf{w}^T S_B \mathbf{w}\end{aligned}$$

where S_B is the between-class variance (known).

Fisher's Linear Discriminant Analysis

Minimizing the within-class variance is equivalent to minimizing the sum of all individual within-class variances. Thus the within class variance is:

$$\begin{aligned}\mathbf{w}^T \Sigma_0 \mathbf{w} + \mathbf{w}^T \Sigma_1 \mathbf{w} &= \mathbf{w}^T (\Sigma_0 + \Sigma_1) \mathbf{w} \\ &= \mathbf{w}^T S_W \mathbf{w}\end{aligned}$$

where S_W is the within-class covariance (known).

Fisher's Linear Discriminant Analysis

To maximize the distance between projected class means and minimize the within-class variance, we can maximize the ratio:

$$\max_{\mathbf{w}} \frac{\mathbf{w}^T S_B \mathbf{w}}{\mathbf{w}^T S_W \mathbf{w}}$$

This is equivalent to finding:

$$\max_{\mathbf{w}} \mathbf{w}^T S_B \mathbf{w}$$

subject to the constraint:

$$\mathbf{w}^T S_W \mathbf{w} = 1$$

Fisher's Linear Discriminant Analysis

To turn this constraint optimization problem into a non-constrained optimization problem, we apply Lagrange multipliers:

$$L(\mathbf{w}, \lambda) = \mathbf{w}^T S_B \mathbf{w} - \lambda(\mathbf{w}^T S_W \mathbf{w} - 1)$$

Differentiating with respect to $\mathbf{w}$ we get:

$$\begin{aligned} \frac{\partial L}{\partial \mathbf{w}} &= 2S_B \mathbf{w} - \lambda 2S_W \mathbf{w} = 0 \\ S_B \mathbf{w} &= \lambda S_W \mathbf{w} \end{aligned}$$

Fisher's Linear Discriminant Analysis

This is a generalized eigenvector problem that is equivalent to (if S_W is not singular):

$$S_W^{-1} S_B \mathbf{w} = \lambda \mathbf{w}$$

where λ and $\mathbf{w}$ are the eigenvalues and eigenvectors of $S_W^{-1} S_B$ respectively.

$\mathbf{w}$ is the eigenvector of $S_W^{-1} S_B$.

Fisher's Linear Discriminant Analysis (A simpler solution)

In fact, for two-classes problems, there exists a simpler solution.

- ▶ Recall that $S_B = (\mu_0 - \mu_1)(\mu_0 - \mu_1)^T$
- ▶ $S_B \mathbf{w} = (\mu_0 - \mu_1)(\mu_0 - \mu_1)^T \mathbf{w}$
- ▶ $(\mu_0 - \mu_1)^T \mathbf{w}$ is a scalar.
- ▶ Therefore $S_B \mathbf{w} \propto (\mu_0 - \mu_1)$. That is, $S_B \mathbf{w}$ is on the same direction as $(\mu_0 - \mu_1)$.
- ▶ And since $S_W^{-1} S_B \mathbf{w} = \lambda \mathbf{w}$, we get:

$$S_W^{-1}(\mu_0 - \mu_1) \propto \mathbf{w}$$

which gives us the direction.

Fisher Discriminant Analysis For Multiple Classes

- ▶ We have defined $\varepsilon(\mathbf{w}) = \frac{\mathbf{w}^T \mathbf{S}_B \mathbf{w}}{\mathbf{w}^T \mathbf{S}_w \mathbf{w}}$ that needs to be maximized.
- ▶ $\mathbf{w}$ is the largest eigen vectors of $\mathbf{S}_w^{-1} \mathbf{S}_B$.
- ▶ For two classes, $\mathbf{w} \propto \mathbf{S}_w^{-1}(\mu_0 - \mu_1)$
- ▶ For k -class problem, Fisher Discriminant Analysis involves $(k - 1)$ discriminant functions.

Fisher Discriminant Analysis For Multiple Classes

Make $W_{d \times (K-1)}$ where each column describes a discriminant. So now, we have to update the two notions we have defined for a 2-class problem, S_B and S_W .

Fisher Discriminant Analysis For Multiple Classes

- ▶ $S_W = \sum_{i=1}^K \Sigma_i$.
- ▶ S_B generalization to multiple classes is not obvious. We will define the total variance S_T as the sum of the within class variance and between classes variance.

$$S_T = S_B + S_W$$

where $S_T = \frac{1}{n} \sum_{i=1}^n (x_i - \mu)(x_i - \mu)^T$

and $\mu = \frac{1}{n} \sum_{i=1}^n x_i$.

So

$$S_B = S_T - S_W$$

Fisher Discriminant Analysis For Multiple Classes

It can be shown that W is the first $(k - 1)$ eigen vectors of $S_W^{-1}S_B$.

Logistic regression

Referred to as Binomial regression in the two class problem.

Goal: Model the probability of being in each class given its predictors by estimating the following functions:

$$P(Y = 1|X = x) = \frac{e^{\beta^T x}}{1 + e^{\beta^T x}}$$

$$P(Y = 0|X = x) = 1 - \frac{e^{\beta^T x}}{1 + e^{\beta^T x}} = \frac{1}{1 + e^{\beta^T x}}$$

Maximum likelihood

Given n data points $\{x_i\}_{i=1}^n$ drawn independently from $p(x; \theta)$, where the form of $p(x)$ is known but θ is unknown, then $\hat{\theta}_{MLE}$ is the Maximum Likelihood Estimate which maximizes the Likelihood of the data.

$$\hat{\theta}_{MLE} = \arg \max_{\theta} \ell(\theta)$$

In this case, we wish to find $\hat{\beta}$ which maximizes $\ell(\beta)$ where

$$\ell(\beta) = \log(L(\beta)) = \sum_{i=1}^n \log(f(x_i; \beta))$$

$$f(x_i; \beta) = \left(\frac{e^{\beta^T x_i}}{1 + e^{\beta^T x_i}} \right)^{y_i} \left(\frac{1}{1 + e^{\beta^T x_i}} \right)^{1 - y_i}$$

Logistic regression

In order to find $\hat{\beta}$ which maximizes $\ell(\beta)$, we set $\frac{\partial \ell}{\partial \beta} = 0$ and solve β .

$$\begin{aligned}\ell(\beta) &= \sum_{i=1}^n \log f(x_i; \beta) \\ &= \sum_{i=1}^n y_i \log \left(\frac{e^{\beta^T x_i}}{1 + e^{\beta^T x_i}} \right) + (1 - y_i) \log \left(\frac{1}{1 + e^{\beta^T x_i}} \right) \\ &= \sum_{i=1}^n y_i \left[\beta^T x_i - \log(1 + e^{\beta^T x_i}) \right] + (1 - y_i) \left[-\log(1 + e^{\beta^T x_i}) \right] \\ &= \sum_{i=1}^n y_i \beta^T x_i - \log(1 + e^{\beta^T x_i})\end{aligned}$$

Logistic regression

$$\frac{\partial \ell}{\partial \beta} = \sum_{i=1}^n y_i x_i^T - \frac{e^{\beta^T x_i}}{1 + e^{\beta^T x_i}} x_i^T$$

We see that $\hat{\beta}$ cannot be found analytically so we can use a numerical method; the Newton Raphson algorithm is widely used:

- 1) Initialize x_0
- 2) $x_{k+1} = x_k - f''(x_k)^{-1} f'(x_k)$
- 3) Repeat until convergence (ie. $|x_{k+1} - x_k| < \epsilon$)

Logistic regression

For convenience, let $p_i = \frac{e^{\beta^T x_i}}{1 + e^{\beta^T x_i}}$ and $1 - p_i = \frac{1}{1 + e^{\beta^T x_i}}$.

Compute the first derivative (Score vector)

$$\frac{\partial \ell}{\partial \beta} = \sum_{i=1}^n (y_i - p_i) x_i^T$$

Compute the second derivative (Hessian matrix)

$$\frac{\partial^2 \ell}{\partial \beta \partial \beta^T} = - \sum_{i=1}^n p_i (1 - p_i) x_i x_i^T$$

Now we can apply the Newton Raphson algorithm to maximize $\ell(\beta)$

$$\beta^{t+1} \leftarrow \beta^t - \left(\frac{\partial^2 \ell}{\partial \beta^t \partial \beta^{tT}} \right)^{-1} \frac{\partial \ell}{\partial \beta^t}$$

Logistic regression

Recalling some matrix algebra, we can convert all summations to matrix operations.

$$\frac{\partial \ell}{\partial \beta} = \mathbf{X}(\mathbf{y} - \mathbf{p})$$

Note: $\mathbf{X}$ is d by n .

$$\frac{\partial^2 \ell}{\partial \beta \partial \beta^T} = -\mathbf{XWX}^T; W_{ii} = p_i(1 - p_i), W_{ij} = 0$$

The Newton-Raphson algorithm can now be expressed as

$$\beta^{t+1} \leftarrow \beta^t + (\mathbf{XWX}^T)^{-1} \mathbf{X}(\mathbf{y} - \mathbf{p})$$

Logistic regression

Alternatively, the algorithm can be expressed as:

$$\begin{aligned}\beta^{t+1} &\leftarrow \beta^t + (\mathbf{X}\mathbf{W}\mathbf{X}^T)^{-1}\mathbf{X}(\mathbf{y} - \mathbf{p}) \\ &\leftarrow (\mathbf{X}\mathbf{W}\mathbf{X}^T)^{-1}[\mathbf{X}\mathbf{W}\mathbf{X}^T\beta^t + \mathbf{X}(\mathbf{y} - \mathbf{p})] \\ &\leftarrow (\mathbf{X}\mathbf{W}\mathbf{X}^T)^{-1}\mathbf{X}\mathbf{W}\mathbf{Z}\end{aligned}$$

where $\mathbf{Z} = \mathbf{X}^T\beta^t + \mathbf{W}^{-1}(\mathbf{y} - \mathbf{p})$

This algorithm is also known as Iteratively Re-weighted Least Squares (IRLS)

$$\beta^{new} \leftarrow \arg \min_{\beta} (\mathbf{Z} - \mathbf{X}\beta)^T \mathbf{W} (\mathbf{Z} - \mathbf{X}\beta)$$

Logistic regression

Note: For a d -dimensional $\mathbf{x}$ this model has d adjustable parameters. By contrast to LDA we have: $2d$ parameters for the means and $d(d+1)/2$ parameters for the covariance matrix. Together with the class priors, LDA gives a total of $d(d+5)/2 + 1$ parameters which grows quadratically in d , in contrast to the linear growth of parameters (d parameters) of logistic regression. For large d , there is a clear advantage for working with the logistic regression model directly.